

# VIBHAS TALLAPALLI

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## EDUCATION

**University of Waterloo**  
BASc in Nanotechnology Engineering

Waterloo, ON  
Sep 2025 – Present

## TECHNICAL SKILLS

**Software:** Python, C++, C, JavaScript, MATLAB, Arduino C++, SQL  
**Libraries/Frameworks:** Flask, Playwright, pandas, Streamlit, Plotly, Chart.js, Tkinter, Matplotlib  
**Data/Databases:** SQLite, normalized schema design, ETL pipelines, real time data logging  
**Quantitative/Finance:** Black Scholes, Greeks (Delta, Gamma, Theta, Vega), Monte Carlo, Sharpe ratio, backtesting, PID control  
**Engineering/Hardware:** SolidWorks, FEA, GD&T, Tolerance, MATLAB simulation, Embedded Systems, Sensor Integration (PPG, IMU)

## EXPERIENCE

**Linear Automation Inc.** | SQL, Python, MATLAB, FEA, SolidWorks  
Mechanical Engineer Co-op

Barrie, ON  
Apr 2026 – Present

- Designed and stress-tested **5+ structural Honda vehicle frame components** in **SolidWorks** using **MATLAB** and **FEA** across **7+ iterations**, running parameter sweeps across wall thickness and load cases to isolate stress concentration failures at oval cross-sections, then redesigning geometry and upgrading material (**AI 5000→6000 series**) to cut component weight by **30%** while maintaining **300+ lb** structural load requirements.
- Queried the **manufacturing execution system** monthly using **SQL**, filtering production data by machine ID, shift, and rejection code to surface defect trends across production runs, identified one machine driving a disproportionate share of night-shift rejects after consistent monitoring, escalated findings to the floor supervisor, and the resulting calibration fix reduced the plant-wide rejection rate by **22%**.
- Built a **Python** script that reads **SolidWorks** drawing feature types and auto-assigns decimal precision per internal **GD&T** standards, eliminating manual precision decisions across **20+ engineering drawings** and reducing decimal precision review comments by **90%** across the team.

**Biomechatronics Engineering Team** | Embedded Systems, C/C++, PPG, IMU, PID Control  
Mechatronics Prototyping Engineer | Design Team

Waterloo, ON  
Sep 2025 – Present

- Designed thigh/shin strap mount designs for a **lower-body exoskeleton**, building **4+ prototypes** with 3D-printed fixtures to improve comfort, cut slip, and eliminate frame interference while maintaining stability during **walking trials** across 4 contact points over repeated wear tests.
- Integrated **PPG heart-rate sensing**, **step tracking**, vibration alerts, and **GPS/notifications** into the exoskeleton strap for prototypes and final build.
- Built the strap sensor data pipeline (**PPG + IMU**), applying filtering and timestamped logging in **C/C++** to generate clean sessions for **PID** tuning.

**IntegraYouth** | Outreach, Public Speaking, Leadership, Event Planning  
Secretary | Blog Writer | Executive

Toronto, ON  
June 2024 – Present

- Wrote **50+ blog posts** on programs and impact, reaching **1,000+ views** and improving outreach through consistent publishing and topic planning.
- Ran Secretary-team operations and supported event logistics, communications, and scheduling for **5+ events** with **200+ attendees**.

## PROJECTS

**Algorithmic Crypto Trading Backtester & Portfolio Analyzer** | Python, pandas, Streamlit, SQLite, Plotly

April 2026

- Built a **full-stack algorithmic crypto trading backtester** in **Python** (**pandas**, **Streamlit**, **SQLite**, **Plotly**) implementing **MA Crossover**, **RSI Mean Reversion**, and **Volume Breakout** strategies across **3 years of OHLCV data**, computing **Sharpe ratio**, **max drawdown**, **win rate**, and **profit factor** across every single backtest run to pressure-test **12 months** of personal paper trading returns against real historical market data.
- Identified a **58% win rate** masking a **1.8x risk/reward ratio** through detailed backtesting analysis, restructuring position sizing by cutting losers faster and extending winners until average win reached **2.1x average loss** at **52% win rate** turning a consistently net-negative strategy profitable.
- Designed a **3 table normalized SQLite schema** with a **SQL Explorer** and **Portfolio Analyzer**, **correlation matrices** across **10 crypto assets**.

**Black-Scholes Live Options Pricing Dashboard** | C++, JavaScript, Chart.js, SQLite

May 2026

- Implemented the **Black-Scholes pricing engine** in **C++** from scratch for European call and put options, pricing **5 simultaneous equities** (**AAPL**, **TSLA**, **NVDA**, **AMZN**, **SPY**) with a custom **CND polynomial approximation** accurate to **6 decimal places** using **zero external math libraries**.
- Computed all **4 Greeks** (**Delta**, **Gamma**, **Theta**, **Vega**) in real time for live hedging analysis, recalculating **8 output values** every **2 seconds** across **1,800 price ticks per hour** with **±0.5%** simulated market microstructure noise and **±1%** volatility surface drift every 10 seconds continuously.

**Job Application Automation Bot** | Python, Flask, Playwright, SQLite, Claude AI

May 2026

- Built a **full-stack job application automation platform** in **Python** (**Flask**, **Playwright**, **SQLite**) that scrapes LinkedIn job postings, tailors LaTeX resumes via **Claude AI** for ATS optimization, and auto-submits applications end to end, cutting per-application time by **97%**.
- Reduced per-application time from **15 minutes to 30 seconds** (**97% reduction**) while maintaining a **10% interview rate** across submissions.
- Designed a **4-table normalized SQLite schema** (jobs, applications, resumes, logs) tracking **6 application stages** with **applicant count filtering** to target low-competition postings, surfaced through a **real-time Flask dashboard** with **2-second live polling** and status counts across **4 stat cards**.

**Electric Bike Control System Simulation** | PID Control, C++, Physics Modelling, Python (Matplotlib)

Jan 2026

- Built a **C++/Python simulation** of an e-bike speed control system, testing **50+ rider load conditions** and analyzing **100+ controller responses** to ensure the bike held a steady target speed while maximizing battery efficiency or feeling unstable when conditions change.
- Implemented and tuned a **PID speed controller** with throttle saturation and anti-windup (integral clamping), reducing **15% overshoot reduction**, **20% settling time improvement**, and improving simulated energy efficiency by **40%** under changing load/terrain.